Devroye–Taylor Rejection Sampling for Univariate Log-Concave Distributions

Abstract

An elegant construction of Devroye shows that univariate log-concave distributions can be sampled exactly by rejection sampling using a three-piece exponential envelope. Given the unnormalised density, this envelope can be constructed by locating the mode of the density, as well as a pair of abscissae located on either side of the mode, at which the density is smaller than its modal value by a fixed constant factor.

We study a simplified 'Devroye–Taylor' method in which these abscissae are not computed exactly, but are instead approximated by making a Taylor approximation to the negative log-density (potential) around the mode. We give explicit acceptance bounds under a relative curvature assumption on the potential, a common structural result covering a range of familiar distributions, and an explicit example showing why some control of curvature variation is necessary.

1 Univariate Log-Concave Sampling à la Devroye

Let X be a random variable of interest with density proportional to $\exp(-V(x))$. Assume that $V : \mathbb{R} \to \mathbb{R}$ is differentiable and convex, that e^{-V} is integrable, and that V has a unique minimizer $x_\star$. We assume access to V and its derivatives but not necessarily to its normalizing constant. For the introductory tangent construction, translate the mode to zero and subtract its height, so that $V(0) = V'(0) = 0$.

The normalizing constant associated with a potential U is denoted by $Z_U = \int_{\mathbb{R}} e^{-U(x)}\,dx$. If $\underline{V} \le V$ is a lower bound on the potential for which $Z_{\underline{V}} < \infty$, define the unnormalized proposal $\gamma(x) = e^{-\underline{V}(x)}$ and the proposal density $q(x) = \gamma(x)/Z_{\underline{V}}$. Drawing a proposal from q and accepting it with probability $e^{\underline{V}(x)-V(x)}$ gives an exact draw from the target density e^{-V}/Z_V. The marginal acceptance probability and expected number of proposals are, respectively, $\alpha(V, \underline{V}) = Z_V/Z_{\underline{V}}$ and $1/\alpha(V, \underline{V})$.

Convexity gives a simple family of lower bounds: every tangent line to V lies below its graph. Choose arbitrary tangent points $u_- < 0 < u_+$ and set

$$v_\pm = V(u_\pm), \qquad p_- = -V'(u_-) > 0, \quad p_+ = V'(u_+) > 0, \qquad L = u_- + \frac{v_-}{p_-}, \quad R = u_+ - \frac{v_+}{p_+}.$$

The tangent inequality at the mode gives $L \le 0 \le R$. The corresponding piecewise-linear minorant and its proposal are

$$\underline{V}(x) = \max\{0, p_-(L - x), p_+(x - R)\}, \qquad \gamma(x) = e^{-\underline{V}(x)}, \qquad q(x) = \frac{\gamma(x)}{Z_{\underline{V}}}. \tag{1}$$

Its normalizing constant is

$$Z_{\underline{V}} = (R - L) + \frac{1}{p_-} + \frac{1}{p_+} = (u_+ - u_-) + \frac{1 - v_-}{p_-} + \frac{1 - v_+}{p_+}. \tag{2}$$

This construction will be used twice, with different choices of the tangent points.

Devroye's original construction chooses the tangent points by computing where the potential exceeds its minimal value by exactly one. Remarkably, this is sufficient to prove a universal acceptance bound [2].

Proposition 1. *Suppose that $V(u_-) = V(u_+) = 1$ for $u_- < 0 < u_+$. Then the proposal defined by (1) satisfies $\alpha(V, \underline{V}) \geq 1 - e^{-1}$. Consequently, the mean number of proposals is at most $e/(e-1)$.*

Proof. Here $v_- = v_+ = 1$, so the general area identity gives $Z_{\underline{V}} = u_+ - u_-$. By convexity, $V(x) \leq x/u_+$ for $0 \leq x \leq u_+$ and $V(x) \leq x/u_-$ for $u_- \leq x \leq 0$. We can hence bound

$$Z_V \geq \int_{u_-}^{0} e^{-x/u_-}\, dx + \int_{0}^{u_+} e^{-x/u_+}\, dx = (u_+ - u_-)(1 - e^{-1}).$$

Dividing by $Z_{\underline{V}} = u_+ - u_-$ proves the claim. □

This guarantee is independent of any other feature of the target. Its practical cost is that computing the abscissae u_-, u_+ typically requires iterative root-finding. Our next step is to replace that calculation with a one-shot Taylor approximation, while retaining the same tangent-envelope construction and hence the exactness of the sampler.

2 Taylor Approximations to the Abscissae

From now on, write $x_\star$ for the minimizer of the transformed potential V (equivalently, the mode of the density) and assume that it has there positive curvature, i.e. $\kappa = V''(x_\star) > 0$. The quadratic approximation

$$V(x_\star + u) - V(x_\star) \approx \frac{\kappa u^2}{2}$$

suggests that $u_\pm$ might be located close to $x_\star \pm \sqrt{2/\kappa}$. We then compute the exact tangents at these points, so that the resulting rejection proposal remains valid regardless of the quality of this guess.

Standardize by $Z = \sqrt{\kappa}(X - x_\star)$ and define

$$W(z) = V\left(x_\star + \frac{z}{\sqrt{\kappa}}\right) - V(x_\star), \qquad r = \sqrt{2}. \tag{3}$$

Then $W(0) = W'(0) = 0$ and $W''(0) = 1$, and the tangent points are $u_- = -r$ and $u_+ = r$. Write

$$v_\pm = W(\pm r), \qquad p_- = -W'(-r), \quad p_+ = W'(r), \qquad L = -r + \frac{v_-}{p_-}, \quad R = r - \frac{v_+}{p_+}.$$

Applying the construction above to W gives $\underline{W}(z) = \max\{0, p_-(L - z), p_+(z - R)\}$, with unnormalized proposal $\gamma(z) = e^{-\underline{W}(z)}$ and density $q(z) = \gamma(z)/Z_{\underline{W}}$. The area identity and acceptance probability then become

$$Z_{\underline{W}} = 2r + \frac{1 - v_-}{p_-} + \frac{1 - v_+}{p_+}, \qquad \alpha(W, \underline{W}) = \frac{Z_W}{Z_{\underline{W}}}. \tag{4}$$

Although individual terms $1 - v_\pm$ may be negative, the normalising constant $Z_{\underline{W}}$ is always strictly positive. The original and standardized target normalizers are related by $Z_W = \sqrt{\kappa}\, e^{V(x_\star)} Z_V$.

The sampler. Compute $x_\star, \kappa, L, R, p_-, p_+$ once. For each proposal:

1. Choose the left tail, cap, or right tail with probabilities proportional to $1/p_-$, $R - L$, and $1/p_+$.
2. If a tail is chosen, then draw $E \sim \mathrm{Exp}(1)$ and set $Z = L - E/p_-$ or $Z = R + E/p_+$. If instead the cap is chosen, then draw Z uniformly from $[L, R]$.
3. Independently draw $U \sim \mathrm{Unif}(0, 1)$. Accept if $\log U \le \underline{W}(Z) - W(Z)$; otherwise repeat.
4. Return $X = x_\star + Z/\sqrt{\kappa}$.

For a Gaussian target the quadratic 'guess' is exact, so that $v_\pm = 1$, $p_\pm = r$, and hence $\alpha(W, \underline{W}) = \sqrt{\pi}/2 \approx 0.886$.

3 Self-Bounding Curvature and Efficiency

A natural first attempt is to control the absolute third derivative, making the quadratic approximation uniformly accurate near the mode. This already gives a useful, if somewhat restrictive, guarantee.

Proposition 2. *Let $W \in C^3(\mathbb{R})$ be convex, with $W(0) = W'(0) = 0$ and $W''(0) = 1$. Suppose $\sup_{z\in\mathbb{R}} |W'''(z)| \le \delta < r = \sqrt{2}$. Then the Devroye–Taylor acceptance probability satisfies*

$$\alpha(W, \underline{W}) \ge \frac{2\displaystyle\int_0^\infty \exp\{-z^2/2 - \delta z^3/6\}\, dz}{2r + \dfrac{2r\delta}{3(r-\delta)}}. \tag{5}$$

Proof. Taylor's theorem yields $|W(z) - z^2/2| \le \delta|z|^3/6$, $p_\pm \ge r - \delta$ and $|v_\pm - 1| \le r\delta/3$. Use the first inequality to bound Z_W from below, and the other two in (4) to bound $Z_{\underline{W}}$ from above. □

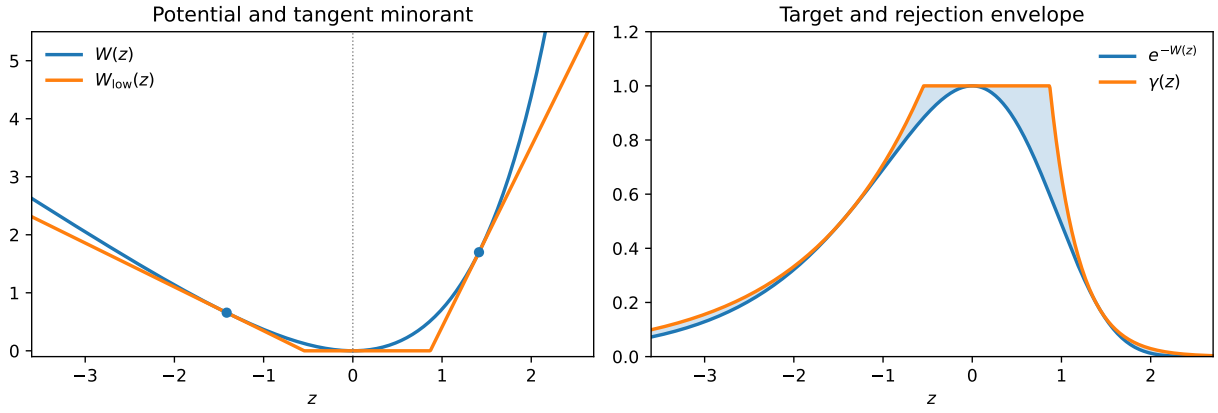


Figure 1: Taylor placement for $W(z) = \mathrm{e}^z - 1 - z$. Left: the quadratic model chooses $\pm\sqrt{2}$, where the true potential has unequal heights. The true tangents and the horizontal line form a lower bound. Right: exponentiating its negative gives a hat above the target. Shaded area is proposal mass that rejection removes. The target is unnormalized in this picture.

In original coordinates, the complexity parameter is given by $\delta = \sup |V'''|/\kappa^{3/2}$. This fails to be useful in the case of, for example, the potential $V(x) = e^x - x$ (which arises in the context of sampling from Gamma distributions), whose third derivative is unbounded. Such potentials may nevertheless have controlled *relative* curvature. We therefore focus on potentials V which satisfy

$$|V'''(x)| \le MV''(x) \qquad (x \in \mathbb{R}), \tag{6}$$

for $M \ge 0$. This generalized self-concordance condition (of exponent two) appears, for example, in analyses of logistic regression [1, 5]. With $D = M/\sqrt{\kappa}$ it is equivalent, after standardization, to the condition $|W'''| \le DW''$.

Theorem 3. *Let $V \in C^3(\mathbb{R})$ satisfy the standing convexity and integrability assumptions, with $\kappa = V''(x_\star) > 0$ and (6). For $i = 0, 1, 2$ define the exponential Taylor remainders*

$$\psi_{i;d}(u) = \sum_{k=0}^{\infty} \frac{d^k u^{i+k}}{(i+k)!}, \qquad d \in \mathbb{R}, \quad u \ge 0.$$

In particular, the series gives $\psi_{i;0}(u) = u^i/i!$. Let $D = M/\sqrt{\kappa} \ge 0$. Writing $[a]_+ = \max(a, 0)$, the acceptance of the Devroye–Taylor sampler satisfies

$$\alpha(W, \underline{W}) \ge \underline{\alpha}(D) := \frac{2\displaystyle\int_0^\infty e^{-\psi_{2;D}(z)}\, dz}{2r + 2\dfrac{[1 - \psi_{2;-D}(r)]_+}{\psi_{1;-D}(r)}} > 0, \qquad r = \sqrt{2}. \tag{7}$$

In particular, $\underline{\alpha}(0) = \sqrt{\pi}/2$.

Proof. The normalized curvature satisfies $W''(0) = 1$ and $|W'''| \le DW''$. Differentiating $e^{Dz}W''(z)$ and $e^{-Dz}W''(z)$ on the positive half-line, and applying the same argument to $W(-z)$, gives

$$\psi_{0;-D}(|z|) \le W''(z) \le \psi_{0;D}(|z|), \qquad z \in \mathbb{R}.$$

Since $\psi'_{i;D} = \psi_{i-1;D}$ for $i = 1, 2$, integrate twice from zero to obtain, for $z \in \mathbb{R}$,

$$\psi_{2;-D}(|z|) \le W(z) \le \psi_{2;D}(|z|).$$

Thus $Z_W \ge 2\int_0^\infty e^{-\psi_{2;D}(z)}\, dz$, $p_\pm \ge \psi_{1;-D}(r)$ and $v_\pm \ge \psi_{2;-D}(r)$. When $1 - v_\pm > 0$, its contribution to (4) is at most $[1 - \psi_{2;-D}(r)]_+/\psi_{1;-D}(r)$; otherwise it is nonpositive. This gives the denominator in (7) and proves the bound. At $D = 0$ the standardized target is Gaussian. □

4 Applications

We develop here some examples of sampling tasks for which the Devroye-Taylor approach is straightforward to apply, and for which efficiency guarantees can be derived painlessly. We will see that these examples are largely accounted for by two building blocks: exponential and log-sum-exp terms. The following result puts them in a common class. The numerical acceptances below are calculated from (4).

Theorem 4. *Suppose that the potential V takes the form*

$$V(x) = \frac{\lambda_0}{2}x^2 + ax + b + \sum_{j=1}^{m} c_j e^{\lambda_j x} + \sum_{k=1}^{n} d_k \log\bigl(c_{0,k} + c_{1,k}e^{\mu_k x}\bigr),$$

where $\lambda_0 \geq 0$, $c_j, d_k > 0$, $c_{0,k} \geq 0$, $c_{1,k} > 0$, V is proper with a unique mode $x_\star$, and $\kappa = V''(x_\star) > 0$. Define

$$M = \max\bigl(\{|\lambda_j|\}_{j=1}^m \cup \{|\mu_k|\}_{k=1}^n \cup \{0\}\bigr), \qquad D = \frac{M}{\sqrt{\kappa}}.$$

Then $|V'''| \leq MV''$ and $\alpha(W, \underline{W}) \geq \underline{\alpha}(D)$.

Proof. For an exponential term, the absolute third derivative is $|\lambda_j|$ times its second derivative. For a log-sum-exp term let

$$\rho_k(x) = \frac{c_{1,k}e^{\mu_k x}}{c_{0,k} + c_{1,k}e^{\mu_k x}}.$$

Its second derivative is $d_k\mu_k^2\rho_k(x)(1-\rho_k(x))$, and its third derivative is this quantity multiplied by $\mu_k(1-2\rho_k(x))$. Since $0 \leq \rho_k \leq 1$, summing the nonnegative second derivatives gives $|V'''| \leq MV''$. The acceptance claim follows from the previous theorem. □

In the examples below, Y will be used to denote the original variable of nominal interest, and $X = T(Y)$ will denote a transformation of said variable which admits a log-concave density.

4.1 Exponential Potentials: Weibull and Gamma

For a positive sum of exponentials plus an affine function, Theorem 4 applies with $M = \max_j |\lambda_j|$. In particular, the Weibull and Gamma transformations below are covered by this single calculation.

Consider a Weibull density with shape $k > 0$ and scale $\lambda > 0$,

$$f_Y(y) = \frac{k}{\lambda}\left(\frac{y}{\lambda}\right)^{k-1} \exp\{-(y/\lambda)^k\}, \qquad y > 0\,.$$

For $X = \log Y$, the change-of-variables formula yields the convex potential

$$V(x) = \mathrm{e}^{k(x-\log\lambda)} - k(x - \log\lambda)\,.$$

The mode occurs at $\log\lambda$, and $\kappa = k^2$. Consequently every shape and scale gives the same standardized potential,

$$W(z) = \mathrm{e}^z - 1 - z\,.$$

This is the example in Figure 1. Here $M = k$, so $D = 1$ for every Weibull law. The substitution $u = \mathrm{e}^z$ gives $Z_W = \mathrm{e}\int_0^\infty \mathrm{e}^{-u}\,du = \mathrm{e}$. Since $v_\pm = \mathrm{e}^{\pm r} - 1 \mp r$ and $p_\pm = \pm(\mathrm{e}^{\pm r} - 1)$, formula (4) gives

$$\alpha(W, \underline{W}) = 0.8893070956\ldots\,.$$

Observe that this improves substantially upon the generic bound $\underline{\alpha}(1) \approx 0.5356$.

The same calculation extends to $Y \sim \mathrm{Gamma}(\alpha, \beta)$, with shape $\alpha > 0$ and rate $\beta > 0$. In logarithmic coordinates, obtain

$$V(x) = \beta\mathrm{e}^x - \alpha x, \qquad x_\star = \log(\alpha/\beta), \qquad \kappa = \alpha, \qquad D = \alpha^{-1/2}\,.$$

Standardization and the substitution $u = \alpha e^{z/\sqrt{\alpha}}$ yield

$$W_\alpha(z) = \alpha\left(e^{z/\sqrt{\alpha}} - 1 - \frac{z}{\sqrt{\alpha}}\right), \qquad Z_{W_\alpha} = \sqrt{\alpha}\, e^{\alpha}\alpha^{-\alpha}\Gamma(\alpha)\,.$$

The rate β disappears after standardization: changing β rescales Y, and hence only translates $X = \log Y$. For $\alpha = 0.1, 1, 10$, the acceptances are approximately $0.900, 0.889, 0.886$. As $\alpha \to \infty$, the distribution and its approximation align with the Gaussian case; one hence has $\alpha(W, \underline{W}) \to \sqrt{\pi}/2$.

4.2 Log-Sum-Exp Potentials: Beta and Spectral Densities

The second building block, $F(s) = \log(c_0 + c_1 e^s)$ with $c_0 \geq 0, c_1 > 0$, also falls under the scope of Theorem 4. Here the associated probabilities $\rho_j(x)$ are as defined in its proof; we use this notation consistently in the Beta and spectral-density examples.

Beta. For $Y \sim \mathrm{Beta}(\alpha, \beta)$ with $\alpha, \beta > 0$, the logit transformation $X = \log\{Y/(1-Y)\}$ gives

$$V(x) = (\alpha+\beta)\log(1+e^x) - \alpha x, \qquad \kappa = \frac{\alpha\beta}{\alpha+\beta}, \qquad D = \sqrt{\frac{1}{\alpha} + \frac{1}{\beta}}\,.$$

As such, even a Beta density that is unbounded at an endpoint (or at both!) still becomes a smooth log-concave target on $\mathbb{R}$ upon transformation. For $(\alpha, \beta) = (1,1), (2,2), (1,2)$, the acceptances are approximately $0.9200, 0.9041, 0.9110$.

Marchenko–Pastur. A specialized exact simulation method for the Marchenko–Pastur law is discussed by Devroye [4]. Here we instead use this law to illustrate the general Taylor–Devroye construction. The continuous part of a Marchenko–Pastur law has density proportional to

$$\frac{\sqrt{(b-y)(y-a)}}{y}, \qquad a < y < b, \qquad a = (1-\sqrt{\theta})^2, \quad b = (1+\sqrt{\theta})^2,$$

where $\theta > 0$. Set $c = a/b$ and use the analogous logit transformation $X = \log\{(Y-a)/(b-Y)\}$. The change-of-variables formula then gives

$$V_c(x) = \log(c + e^x) + 2\log(1+e^x) - \frac{3}{2}x, \qquad 0 \leq c < 1\,. \tag{8}$$

Theorem 4 gives $M = 1$, including when $c = 0$. This time, there is also a lower bound on modal curvature that holds across all parameters. Define

$$\rho_1(x) = \frac{e^x}{c+e^x}, \qquad \rho_2(x) = \frac{e^x}{1+e^x}.$$

The mode equation and curvature are

$$\rho_1(x_\star) + 2\rho_2(x_\star) = \frac{3}{2}, \qquad \kappa = \rho_1(x_\star)(1-\rho_1(x_\star)) + 2\rho_2(x_\star)(1-\rho_2(x_\star)) = \frac{3}{4} - 6\left(\rho_2(x_\star) - \frac{1}{2}\right)^2.$$

Since $0 \leq \rho_1(x_\star) \leq 1$, we have $1/4 \leq \rho_2(x_\star) \leq 3/4$. Hence $3/8 \leq \kappa \leq 3/4$, so $D \leq \sqrt{8/3}$. Theorem 3 proves a uniform positive acceptance bound.

The mode is explicit as $x_\star = \log\left(\frac{1-c+\sqrt{1+34c+c^2}}{6}\right)$, and the normalized density is $K_c \exp(-V_c)$, where $K_c = 2(1+\sqrt{c})^2/\pi$, so $Z_W = \sqrt{\kappa}\, e^{V_c(x_\star)}/K_c$. This gives a scalar formula for acceptance. A numerical search over $0 \leq c \leq 1$, including the limit at $c = 1$, finds a minimum near $c = 0.00354$ with acceptance about 0.883.

Wachter / Free Jacobi. For the continuous density proportional to $\sqrt{(b-y)(y-a)}/\{y(1-y)\}$ with $0 < a < b < 1$, the same change-of-variables strategy leads to the potential

$$V(x) = \log(1+\mathrm{e}^x) + \log(a+b\mathrm{e}^x) + \log\{(1-a)+(1-b)\mathrm{e}^x\} - \frac{3}{2}x\,.$$

Again $M = 1$. Define $\rho_1(x) = \mathrm{e}^x/(1+\mathrm{e}^x)$, $\rho_2(x) = b\mathrm{e}^x/(a+b\mathrm{e}^x)$ and $\rho_3(x) = (1-b)\mathrm{e}^x/((1-a)+(1-b)\mathrm{e}^x)$. At the mode these satisfy $\sum_j \rho_j(x_\star) = 3/2$, and $\kappa = 3/2 - \sum_j \rho_j(x_\star)^2$. Under $0 \le \rho_j(x_\star) \le 1$, the sum of squares is at most $1 + (1/2)^2 = 5/4$, as moving mass from a smaller to a larger coordinate increases the sum until all but at most one coordinate are endpoints. Hence $\kappa \ge 1/4$, $D \le 2$, and $\alpha(W, \underline{W}) \ge \underline{\alpha}(2)$.

4.3 Log-Cosh Potentials, Three Ways

The potential $V(x) = \nu \log\cosh x$, with $\nu > 0$, unifies several more examples. Since $\log\cosh x = \log(1+e^{2x}) - x - \log 2$, it is also covered by Theorem 4. Here $\kappa = \nu$ and $V'''/V'' = -2\tanh x$, so $M = 2$ and $D = 2/\sqrt{\nu}$. With some work, one can see that

1. For the Student t_ν distribution, the transformation $X = \operatorname{arsinh}(Y/\sqrt{\nu})$ falls into this family, with parameter ν.
2. For the Beta$(\nu/2, \nu/2)$ distribution, the transformation $X = \frac{1}{2}\log\frac{Y}{1-Y}$ does the same, also with parameter ν.
3. For the symmetric distribution on $(-1, 1)$ with density proportional to $(1-y^2)^\lambda$ for some $\lambda > -1$, the transformation $X = \operatorname{artanh} Y$ also falls into this family, with parameter $\nu = 2\,(\lambda + 1)$.

In particular, the semicircle law corresponds to $\lambda = 1/2$ and $\nu = 3$.

Writing B for the beta function, one computes the relevant auxiliary quantities as

$$Z_{W_\nu} = \sqrt{\nu}\,\mathrm{B}(1/2, \nu/2), \qquad v_\pm = \nu \log\cosh(r/\sqrt{\nu}), \qquad p_\pm = \sqrt{\nu}\tanh(r/\sqrt{\nu})\,.$$

Formula (4) gives acceptance rates of approximately $0.944, 0.920, 0.910$ for $\nu = 1, 2, 3$.

4.4 Sliced Distributions in GLMs

For logistic and Poisson regression with a Gaussian prior, the posterior potential restricted to a line has, up to an additive constant, the form

$$V(x) = \frac{\lambda}{2}x^2 + \ell x + \sum_i \{F(a_i x + b_i) - y_i(a_i x + b_i)\}, \qquad F(s) = \begin{cases} \log(1+\mathrm{e}^s), & \text{logistic,} \\ \mathrm{e}^s, & \text{Poisson.} \end{cases}$$

Here $\lambda \ge 0$, with binary responses in the logistic case and nonnegative integer counts in the Poisson case. A flat prior corresponds to $\lambda = \ell = 0$, provided that the resulting conditional is proper. This is a special case of Theorem 4 with $M = \max_i |a_i|$. For a proper conditional with modal curvature $\kappa > 0$, the resulting acceptance bound is $\alpha(W, \underline{W}) \ge \underline{\alpha}(M/\sqrt{\kappa})$. In particular, a positive quadratic coefficient $\lambda > 0$ guarantees a proper, strongly convex conditional.

5 A Degenerate Limit

Theorem 3 provides a guarantee of uniform efficiency at fixed D which degrades as D diverges. We demonstrate here that this is not just a failure of the bound, and that the Taylor approximation can occasionally behave rather poorly. The generalized inverse Gaussian (GIG) family supplies a concrete example to this effect; see Devroye [3] for specialized sampling methods for this family. For

$$f_Y(y) \propto y^{p-1} \exp\left\{-\frac{1}{2}\left(ay + \frac{b}{y}\right)\right\}, \qquad a, b > 0, \quad p \in \mathbb{R},$$

the logarithmic transformation gives the convex potential

$$V(x) = \frac{a}{2}\mathrm{e}^{x} + \frac{b}{2}\mathrm{e}^{-x} - px, \qquad \kappa = \sqrt{p^2 + ab}, \qquad M = 1\,,$$

with $D = (p^2 + ab)^{-1/4}$.

Take now $p = 0$ and $a = b = \varepsilon \downarrow 0$. After subtracting the modal value, we obtain

$$V_\varepsilon(x) = \varepsilon(\cosh x - 1), \qquad W_\varepsilon(z) = \varepsilon\{\cosh(z/\sqrt{\varepsilon}) - 1\}\,.$$

Here $\kappa = \varepsilon$ and $D = \varepsilon^{-1/2} \to \infty$. The exact abscissae advocated for in Devroye's construction then satisfy $u_\pm = \pm\operatorname{arcosh}(1 + 1/\varepsilon)$, with $u_+ \sim \log(2/\varepsilon)$, whereas the Taylor approximation suggests to instead seek them at $\pm\sqrt{2/\varepsilon}$. Thus the guessed abscissae lie much too far from the mode: the potential grows exponentially away from the mode, whereas its quadratic approximation predicts a much larger distance before reaching height one.

Proposition 5. *For this symmetric GIG sequence, the Devroye–Taylor acceptance satisfies*

$$\alpha(W_\varepsilon, \underline{W}_\varepsilon) \sim \frac{\sqrt{\varepsilon}}{\sqrt{2}} \log(2/\varepsilon) \longrightarrow 0\,.$$

Proof. Symmetry gives $v_\pm = \varepsilon\{\cosh(r/\sqrt{\varepsilon}) - 1\}$ and $p_\pm = \sqrt{\varepsilon}\sinh(r/\sqrt{\varepsilon})$. Therefore $v_\pm/p_\pm \sim \sqrt{\varepsilon}$ and $1/p_\pm \to 0$, so (4) gives $Z_{\underline{W}_\varepsilon} \to 2r$. For the mass, change variables back to y, and write

$$Z_{W_\varepsilon} = 2\sqrt{\varepsilon}\,\mathrm{e}^{\varepsilon} I_\varepsilon, \qquad I_\varepsilon = \int_0^\infty \mathrm{e}^{-\varepsilon\cosh x}\,dx\,.$$

Put $T = \log(1/\varepsilon)$ and fix $0 < \eta < 1$. The integrand tends uniformly to one on $[0, (1-\eta)T]$, giving $I_\varepsilon \geq (1-\eta)T\{1 - o(1)\}$. Its integral up to $(1+\eta)T$ is at most $(1+\eta)T$. Since $\cosh x \geq \mathrm{e}^x/2$, the remaining tail is at most

$$\int_{(1+\eta)T}^\infty \mathrm{e}^{-\varepsilon \mathrm{e}^x/2}\,dx = \int_{\varepsilon^{-\eta}/2}^\infty \frac{\mathrm{e}^{-u}}{u}\,du = o(1)\,.$$

Letting $\eta \downarrow 0$ proves $I_\varepsilon \sim T \sim \log(2/\varepsilon)$. Divide the resulting mass asymptotic by $Z_{\underline{W}} \to 2\sqrt{2}$. □

6 Conclusion

The construction separates various considerations cleanly. Convexity allows the construction of a valid sampler, careful construction of a rejection envelope allows for uniform efficiency, Taylor approximation allows for a cheaper construction of an envelope which remains valid, and control of the potential curvature ensures that this simplified envelope remains performant. The examples show that this cheap quadratic guess often works well, while the GIG limit shows that the Taylor-placement sampler cannot have a parameter-free efficiency guarantee over all one-dimensional log-concave targets.

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